Halward Inspection Services Ltd.

Inspection Report

Site: Hamilton Plant

No.	Status	Date
HIS-2013-0402	Issued	2013-09-09

UT Thickness Survey - Regenerator Overhead and Reflux

Scope and Responsibilities

This report details the 3-year regenerator overhead and reflux survey conducted at the Hamilton Plant within the ATU-100. The scope includes ultrasonic thickness measurements of lines 3-RV-101, 2-RV-102, 3-RF-101, and 2-RF-102 to monitor for internal thinning and assess the current corrosion rates. Kevin Doucette served as the lead UT technician, responsible for data acquisition and the preliminary evaluation of results against established retirement limits.

Inspection activities were performed under QSCP-NDT-001 Rev 2 and HIS-UT-002 Rev 1. These procedures ensure compliance with the technical requirements of ASTM E797 and the in-service inspection mandates of API 570. The survey follows the specific requirements of QSCP-IP-ATU100 Rev 2, which incorporates the CUI program protocols for insulated lines operating within the CUI temperature range.

Inspection Methodology

Measurements were obtained using digital ultrasonic thickness gauges calibrated in accordance with ASTM E797. For the carbon steel regenerator overhead lines 3-RV-101 and 2-RV-102, surface preparation was performed at established CMLs to ensure accurate coupling. The stainless steel reflux lines 3-RF-101 and 2-RF-102 were inspected using high-resolution transducers suitable for the material thickness and design temperatures of 104 °C.

In alignment with QSCP-NDT-001 Rev 2, the inspection included the assessment of insulation windows and profile radiography where required by the CUI program. This methodology addresses the revised requirements for profile RT or insulation removal every 5 years for lines identified in QSCP-IP-ATU100 Rev 2 as susceptible to external corrosion under insulation.

Data Analysis and Comparison

Current thickness readings were compared against the baseline and historical data recorded in HIS-2010-0318. For line 3-RV-101, which has a t-min of 3.56 mm, and line 2-RV-102, with a t-min of 2.79 mm, the short-term corrosion rates were calculated using the 3-year interval since the previous survey. The reflux lines 3-RF-101 and 2-RF-102 were evaluated against their respective t-min values of 2.41 mm and 2.16 mm.

Corrosion rates were determined by selecting the larger of the long-term and short-term rates as per site rules. Remaining life calculations utilized these rates to establish the next inspection date. Because the regenerator overhead circuit is classified as Class 1, the thickness interval is capped at 3 years, while the Class 2 reflux lines are subject to a maximum 10-year thickness and 5-year external visual interval.

Summary of Findings

All measured thicknesses for 3-RV-101, 2-RV-102, 3-RF-101, and 2-RF-102 remain above the specified retirement thickness (t-min). No NCRs were raised during this survey as no readings fell below the allowable limits defined in the line facts. The data indicates that the corrosion rates are within the predicted ranges established during the HIS-2010-0318 inspection.

The next thickness reading for the ATU-100 regenerator overhead lines is due in 2016, maintaining the 3-year interval required for this circuit. The reflux lines will continue on their assigned Class 2 schedule, with the next external visual inspection due within 5 years. Kevin Doucette has verified that all CMLs remain accessible and that the insulation reinstatement meets the requirements of QSCP-NDT-001 Rev 2.

Thickness readings, measured 2013-09-09

Line 3-RV-101 | Regenerator overhead | A106 Gr.B | t_nom 7.62 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RV-101-01	7.62	3.56	2010-09-13	6.78	6.49	0.063	0.077	38.1	2016-09-09	
3-RV-101-02	7.62	3.56	2010-09-13	6.59	6.44	0.065	0.075	38.3	2016-09-09	
3-RV-101-03	7.62	3.56	2010-09-13	6.16	5.80	0.098	0.127	17.6	2016-09-09	
3-RV-101-04	7.62	3.56	2010-09-13	6.43	6.13	0.083	0.089	29.0	2016-09-09	
3-RV-101-05	7.62	3.56	2010-09-13	6.63	6.48	0.068	0.065	43.0	2016-09-09	

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RV-101-06	7.62	3.56	2010-09-13	6.43	6.11	0.085	0.100	25.4	2016-09-09	
3-RV-101-07	7.62	3.56	2010-09-13	6.20	5.73	0.157	0.157	13.8	2016-09-09	
3-RV-101-08	7.62	3.56	2010-09-13	6.71	6.36	0.117	0.117	23.9	2016-09-09	

Line 2-RV-102 | Regenerator overhead | A106 Gr.B | t_nom 5.54 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RV-102-01	5.54	2.79	2010-09-13	4.55	4.49	0.059	0.057	28.6	2016-09-09	
2-RV-102-02	5.54	2.79	2010-09-13	4.60	4.29	0.074	0.064	20.2	2016-09-09	
2-RV-102-03	5.54	2.79	2010-09-13	4.84	4.68	0.046	0.055	34.2	2016-09-09	
2-RV-102-04	5.54	2.79	2010-09-13	4.84	4.81	0.044	0.052	38.9	2016-09-09	
2-RV-102-05	5.54	2.79	2010-09-13	5.04	4.96	0.027	0.027	>50	2016-09-09	
2-RV-102-06	5.54	2.79	2010-09-13	4.82	4.55	0.090	0.090	19.5	2016-09-09	

Line 3-RF-101 | Reflux | A312 TP316 | t_nom 5.49 mm | t-min 2.41 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RF-101-01	5.49	2.41	2006-02-24	5.48	5.39	0.005	0.012	>50	2023-09-09	
3-RF-101-02	5.49	2.41	2006-02-24	5.37	5.34	0.006	0.004	>50	2023-09-09	
3-RF-101-03	5.49	2.41	2006-02-24	5.45	5.45	0.000	0.000	>50	2023-09-09	
3-RF-101-04	5.49	2.41	2006-02-24	5.33	5.33	0.007	0.000	>50	2023-09-09	
3-RF-101-05	5.49	2.41	2006-02-24	5.40	5.26	0.011	0.019	>50	2023-09-09	
3-RF-101-06	5.49	2.41	2006-02-24	5.40	5.25	0.015	0.020	>50	2023-09-09	
3-RF-101-07	5.49	2.41	2006-02-24	5.35	5.43	-0.001	-0.011	>50	2023-09-09	
3-RF-101-09	5.49	2.41	—	—	5.12	—	—	—	2023-09-09	

Line 2-RF-102 | Reflux | A312 TP316 | t_nom 3.91 mm | t-min 2.16 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RF-102-01	3.91	2.16	2006-02-24	3.85	3.71	0.012	0.019	>50	2023-09-09	
2-RF-102-02	3.91	2.16	2006-02-24	3.88	3.65	0.013	0.031	48.8	2023-09-09	
2-RF-102-03	3.91	2.16	2006-02-24	3.80	3.76	0.010	0.005	>50	2023-09-09	
2-RF-102-04	3.91	2.16	2006-02-24	3.81	3.80	0.004	0.001	>50	2023-09-09	
2-RF-102-05	3.91	2.16	—	—	3.69	—	—	—	2023-09-09	
2-RF-102-06	3.91	2.16	—	—	3.71	—	—	—	2023-09-09	
2-RF-102-07	3.91	2.16	2006-02-24	3.85	3.82	0.008	0.004	>50	2023-09-09	

Sign-off	Name	Function	Date	Signature
Prepared	Kevin Doucette	lead UT technician, SNT-TC-1A UT Level II	2013-09-09	_____